Loot or Love? A Monte Carlo Comparison of Pirate Shop Token Acquisition and Romantic Partnership Formation Among Fortnite Players

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Abstract

Fortnite's Pirate Shop Access Token, which makes the entire Item Shop free for 14 days, drops from 1 in 137,500 chests. I estimate that a right swipe on a dating app has about a 1 in 97,600 chance of ending in a relationship, a cold social-media message about 1 in 7,300, and meeting a new person face to face about 1 in 23,500. I simulate both pursuits. The token model covers game mode, region, skill-based matchmaking (SBMM), longer queues at higher skill, and fatigue within a session. The relationship model covers in-person meeting, dating apps, direct messages, physical attractiveness, local population density, and the socialising time that gaming takes up. Both models are calibrated to published data, and every assumed parameter is given a prior and propagated through the full simulation. The median solo player needs about 3,240 hours to find the token, and by their 800th match spends more than half of their Fortnite time outside a match. I also derive the survival time that maximises chests per hour in closed form: 5 minutes 49 seconds. In the 26 days left in the season, the token overtakes a girlfriend at about 19 hours of play per week for the average man (90% interval 15 to 46). For men in the bottom fifth for looks it happens at 11 hours, and for the top fifth at 32. Living rurally, not using dating apps and playing squads all make the comparison worse, in different directions.

1 Introduction

On 7 November 2026 Epic Games published the odds of every treasure in its current Battle Royale season [1, 2]. The most valuable item, the Pirate Shop Access Token, appears in a given chest with probability $p = 1/137{,}500$. A figure of 1 in 700,000 had circulated before this and turned out to be wrong [4]. The correction made the token five times more likely, which still left it very unlikely.

Odds per chest are the wrong unit for a human decision. People experience odds in hours, and in the slowly growing suspicion that those hours could have gone somewhere else. For the median player, the obvious "somewhere else" is a social life. This paper asks which is more likely to arrive first: a free Item Shop or a girlfriend.

The question has a respectable ancestor. Backus [5] applied the Drake equation, originally used to estimate the number of alien civilisations in the galaxy, to his own dating prospects and concluded there were 26 suitable women in the United Kingdom. I extend his approach in four ways. I model dating as a process over time rather than a fixed pool. I add the channels most people now actually use, which are dating apps and social media. I let physical attractiveness and geography matter, because they do. And I put the whole thing head to head with a loot table.

2 Data

Token. Epic's table gives $p = 1/137{,}500$ per chest opened in a normal Battle Royale match. The rate is constant except during "Treasure Hours", when it rises by at least 1.5× and has reached 2× [1, 3]. The season ends on 5 December 2026 [2], 26 days from the date of writing. Epic has not said whether a token found by one squad member counts for the others, so the main results are per player.

Time use. The 2025 American Time Use Survey [6] reports that people aged 20 to 24 spend 0.51 hours a day socialising and 1.08 hours gaming or using a computer for leisure. Since 2015, the average time spent socialising across all ages has fallen from 41 to 35 minutes a day.

Fatigue. In League of Legends, win rate and kill/death/assist ratio fall by more than 10% and 8% from the first to the last match of a session [7], and performance declines with the number of consecutive matches played [8].

Relationships. Pew reports that 63% of men and 34% of women aged 18 to 29 are single [9], and that roughly half of single adults are not looking [10]. Wolfinger [12] argues the gender gap is closer to 10 points than 30, so I also report a lower calibration ("GSS", $s = 0.53$). In 2017, 39% of heterosexual couples met online, having overtaken meeting through friends (20%) around 2013 [13].

Online dating. Of adults under 30, 53% have used a dating app or site, and 64% of male users say they have felt insecure about how few messages they got [11]. In a study of Tinder in London, men's right swipes produced a match 0.6% of the time, against 10.5% for women [14]. On OkCupid, women rated 80% of men as worse-looking than average, and the most attractive men received about 11 times as many messages as the least attractive [15].

3 Model

3.1 Token hunting

Each simulated player i has a skill level that sets the survival time SBMM eventually pushes them towards, $T_i^{\infty} = 6\,e^{z_i}$ minutes with $z_i \sim \mathcal{N}(0, 0.35^2)$. In match n their expected survival is

$$\bar{T}_{in} = (1-\phi)^{k_n}\Big[T_i^{\infty} + (T_0 - T_i^{\infty})\,e^{-n/\tau}\Big], \qquad (1)$$

with $T_0 = 14$ minutes for a new account and $\tau = 150$ matches. The factor $(1-\phi)^{k_n}$ is fatigue. Here k_n is the position of the match within a three-hour session, and $\phi = 0.01$ per match, which reproduces the roughly 10% decline in [7]. Actual survival is drawn from a gamma distribution with this mean and shape 2. Chests opened are Poisson with mean

$$\mathbb{E}[c_{in}] = \sigma_m\, C\big(1 - e^{-(1-\phi)^{k_n} T_{in}/\kappa}\big), \qquad (2)$$

where $C = 12$ is the most chests a player opens in a long game and $\kappa = 6$ minutes captures the fact that looting happens early. Tired players loot more slowly too. The share σ_m is the fraction of chests a player gets when teammates compete for them: 1 for solos, 0.85 for duos, 0.75 for trios and 0.65 for squads.

Every match also costs time outside the match itself: returning to the lobby, queueing, loading and the Battle Bus. I model this as

$$O_n = O_0 + \Delta_Q\big(1 - e^{-n/\tau}\big) + \eta\, k_n + q_r, \qquad (3)$$

where $O_0 = 4$ minutes and $\Delta_Q = 1.5$ minutes reflects longer queues as SBMM moves a player into smaller, higher-rated pools. The term $\eta = 0.15$ minutes per match captures the growing pause between games as a session wears on, and q_r (0 to 2 minutes) is extra queue time in lower-population regions. Regions also scale survival, from 0.90 in Brazil to 1.15 in Asia, to reflect how many bots are in a lobby.

To simulate a player's time to the token exactly, I draw the number of chests they will open before finding it, $K_i \sim \text{Geometric}(p)$, and simulate matches until they have opened K_i chests. For the season window I use the lower-variance estimator $\Pr(\text{token}) = \mathbb{E}[1 - e^{-p\sum_n c_{in}}]$, starting from a player with 300 matches of history.

3.2 The girlfriend equation

A man j enters a relationship at annual rate

$$\lambda_j = r_j\big[\underbrace{\ell^{\text{off}}(u_j)\,\lambda_{\text{off}}(g)}_{\text{in person}} + \underbrace{\ell(u_j)\,(A_j\lambda_{\text{app}} + \lambda_{\text{DM}})}_{\text{online}}\big], \quad (4)$$

where g is daily gaming hours, $u_j \sim U(0,1)$ is his looks percentile, A_j indicates whether he uses dating apps (probability 0.53), and r_j is a lognormal factor with mean 1 and log-sd 0.5 that covers personality, confidence and luck.

In person. Following Backus [5], the chance that a new person becomes a partner is

$$\pi = \pi_{\text{F}}\,\pi_{\text{age}}\,\pi_{\text{single}}\,\pi_{\text{look}}\,\pi_{\text{attr}}\,\pi_{\text{compat}}\,\pi_{\text{ask}}\,\pi_{\text{yes}}, \qquad (5)$$

with $\pi_{\text{single}} = 0.34$ and $\pi_{\text{look}} = 0.5$ from Pew, $\pi_{\text{attr}} = 1/20$ and $\pi_{\text{compat}} = 1/10$ from Backus, and the other four set at 0.5, 0.4, 0.5 and 0.5. This gives about 1 in 23,500. New people are met at $E(g) = E_{\text{ctx}} + \beta\max\{0, S_0 - \delta(g - g_0)\}$ per day, where $E_{\text{ctx}} = 0.6$ come from school or work, $\beta = 0.8$ per hour of socialising, S_0 and g_0 are the ATUS averages, and each extra hour of gaming displaces $\delta = 0.2$ hours of socialising. Then $\lambda_{\text{off}}(g) = 365\,M\,E(g)\,\pi$, where M is a calibrated multiplier that I return to below.

Dating apps. Each right swipe matches with probability 0.6% [14]. A match becomes a conversation with probability 0.3, a conversation becomes a date with probability 0.15, and a date becomes a relationship with probability ρ, which is calibrated. An active user swipes 40 times a day, but no one can swipe on people who are not there. The pool of eligible women within range is

$$N_{\text{pool}} = D\,\pi R^2 \times 0.08 \times 0.34 \times 0.3, \qquad (6)$$

where D is population density, R is the search radius, and the factors are the share of the population who are women aged 18 to 29, the share of those who are single, and the share currently active on an app. A user can work through their local pool about once a month, so daily swipes are capped at $N_{\text{pool}}/30$. This cap does not bind in a city of 5,000 people per km^2 (12,800 profiles) but leaves a rural user (20 per km^2, 25 km radius) with 320 profiles and 11 swipes a day.

Social media. He also sends 3 cold messages a week to women he does not know well, with a reply rate of 8%, followed by the same conversation, date and relationship steps.

Looks. Attractiveness multiplies online success by $\ell(u) \propto e^{u \ln 11}$, normalised to mean 1, so the most attractive man does 11 times better than the least attractive [15]. In person I assume looks matter less, with a range of 4.

Calibration. Two parameters, M and ρ, are fixed by two published targets. With a breakup rate of $\mu = 0.35$ per year, the simulated population must reproduce Pew's single share, $\mathbb{E}_j[\mu/(\lambda_j + \mu)] = 0.63$, and online channels must supply 39% of new couples [13]. This gives $M = 10.1$ and $\rho = 0.038$. Two consequences follow. First, a literal reading of the Drake equation ($M = 1$) implies a median wait of 44 years, so repeated contact with people one already knows makes a young man about ten times more likely to find a partner than one-off encounters suggest. Second, because the targets are fixed, many assumptions cancel out. If swipes were half as effective, for example, the calibration would make each date twice as likely to work out.

4 Results

4.1 The odds per attempt

Figure 1 shows the two funnels. A right swipe is worth about 1 in 97,600, which is closer to a Fortnite chest than most users would like to know. A cold message is worth 1 in 7,300, and meeting someone new in person 1 in 23,500. Love beats loot on every channel per attempt. The problem is the number of attempts: a player opens around 30 chests per hour, but meets about one new person per day.

4.2 Token by mode and region

The median solo player finds the token after about 3,240 hours (Table 1, Figure 2). The luckiest tenth need under 480 hours, and the unluckiest tenth need over 10,800. Squads are the slowest way for an individual to play, needing 54% longer than solos, because teammates open chests that would otherwise be yours. If Epic confirmed that a find counts for the whole squad, four players pooling chests would need about 1,170 hours each.

Region matters less than expected. Europe and North America need about 3,200 hours, while Brazil, Asia, Oceania and the Middle East need between 3,310 and 3,550, mostly because of longer queues. Treasure Hours reduce the median to 2,110 hours at 1.5× and 1,580 at 2×. They happen about four hours a week, so a player who only played during them would wait eight to ten years.

Table 1: Hours of play needed to find the token, per player.

Mode	Luckiest 10%	Median	Unluckiest 10%
Solo	480	3,244	10,800
Duo	565	3,821	12,700
Trio	640	4,331	14,378
Squad	740	4,991	16,603

4.3 The lobby problem

Harder lobbies mean shorter games, and shorter games mean a larger share of time spent outside them (Figure 3). A new, rested player spends about 30% of their Fortnite time in lobbies, queues and loading screens. By their 800th match this reaches 48%, and with fatigue 54%. Beyond that point, a typical player spends more of their Fortnite time not playing Fortnite than playing it.

This leads to a decomposition of SBMM's effect at 10 hours a week. Shorter survival alone *raises* the season probability by about 6%, because chests are opened early and a player who dies at minute six lands on a fresh map sooner. Longer queues at higher skill take away about 9%. The net effect of SBMM is −4% (0.82% against 0.85% without SBMM). Fatigue costs another 11%, so the best way to hunt is in short, rested sessions.

4.4 The optimal time to die

The SBMM result has a cleaner form if fatigue and randomness are switched off. A player who survives exactly T minutes in every match, with overhead O, opens chests at a rate of

$$r(T) = \frac{60\,C\,(1 - e^{-T/\kappa})}{T + O} \text{ per hour.} \tag{7}$$

Proposition 1. *For $O, \kappa > 0$, $r(T)$ has a unique maximum at*

$$T^* = -\kappa\Big[W_{-1}\big(-e^{-(1+O/\kappa)}\big) + 1\Big] - O, \tag{8}$$

where W_{-1} is the lower branch of the Lambert W function.

Proof. Setting $r'(T) = 0$ gives $e^{T/\kappa} = 1 + (T + O)/\kappa$. Write $x = 1 + (T + O)/\kappa$, so that $e^{x-1-O/\kappa} = x$, or $-xe^{-x} = -e^{-(1+O/\kappa)}$. The argument lies in $(-1/e, 0)$, so there are two real solutions. The principal branch gives $x < 1$, which implies $T < -O$, so the feasible root is $x = -W_{-1}(-e^{-(1+O/\kappa)})$. The numerator of $r'(T)$ is decreasing in T, so the critical point is a maximum. □

With $O = 4$ and $\kappa = 6$, $T^* = 5.82$ minutes, or 5 minutes 49 seconds, at 45.5 chests per hour. The optimum grows only slowly with overhead: 4.3 minutes at $O = 2$ and 7.3 at $O = 7$. Figure 4 checks the formula against

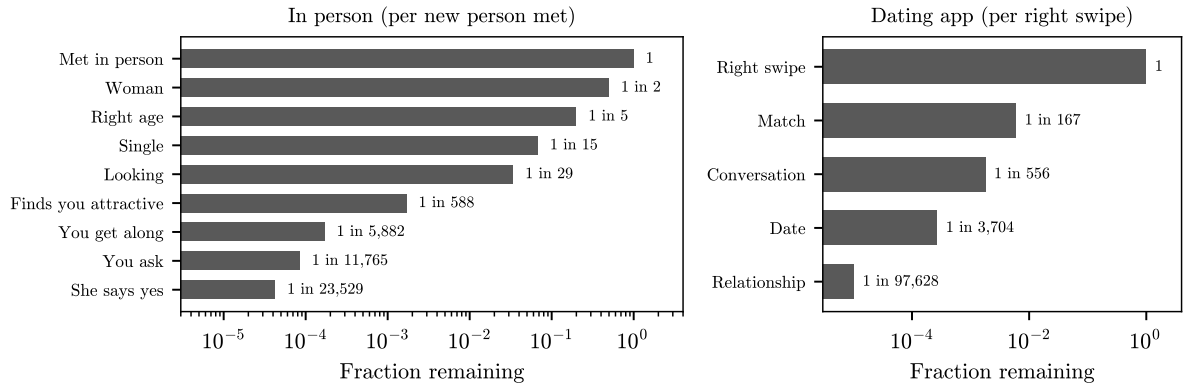


Figure 1: The two funnels. Cumulative fraction surviving each filter, per new person met in person (left) and per right swipe (right). For comparison, the token is 1 in 137,500 per chest.

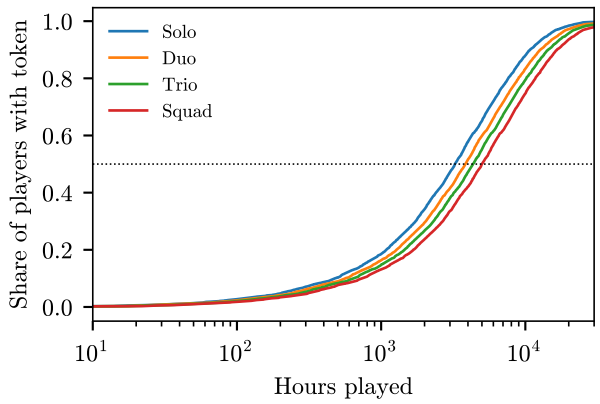


Figure 2: Share of simulated players who have found the token, by hours played and mode.

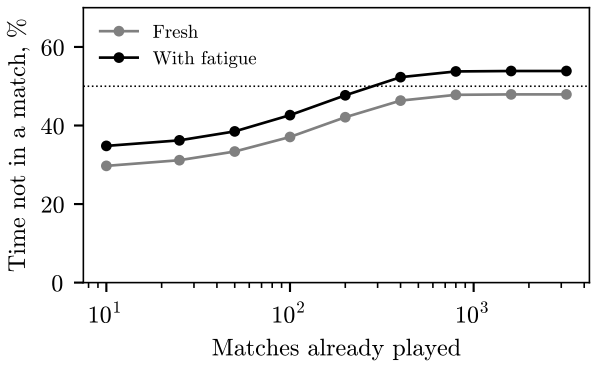


Figure 3: Share of total Fortnite time spent outside a match (lobby, queue, loading, Battle Bus), by matches already played.

the full simulation run at fixed survival times. Mathematically, a player who has looted the first circle and is not about to win has little reason to stay. I do not recommend acting on this, but I suspect many players already do.

4.5 Looks, location and apps

Looks matter a great deal. Over the next 26 days, a man at 10 hours a week has a 1.8% chance of a relationship on average. The figure is 0.9% for the bottom fifth by looks and 3.0% for the top fifth. Dating app users have nearly double the chance of non-users (2.2% against 1.2%). Geography matters through the app pool. A city or suburb gives 1.7%, a small town 1.5% and a rural area 1.3%, where the user runs out of new profiles to swipe on.

One finding has practical value. A player at 25 hours a week who spends one hour a day of queue time swiping, within the limits of his local pool, raises his season probability from 1.4% to 1.9%. That is higher than a 10-hour-a-week player who does not swipe. Queue time, previously wasted, becomes the single most efficient use of a Fortnite player's evening.

4.6 The season race

Figure 5 puts both sides together. A man who does not play at all has a 2.1% chance of a relationship before 5 December and no chance of the token. As weekly hours rise, the token's chance grows roughly linearly while the relationship's falls, as gaming displaces socialising. For the average man the curves cross at about 19 hours a week (27 under the GSS calibration). For the bottom fifth by looks they cross at 11 hours, and for the top fifth at 32. At 56 hours a week the token is about three times more likely than a girlfriend (4.4% against 1.4%). Beyond about 25 hours a week the relationship curve flattens, because there is no free socialising left to lose and the remaining chances come from school, work and the phone.

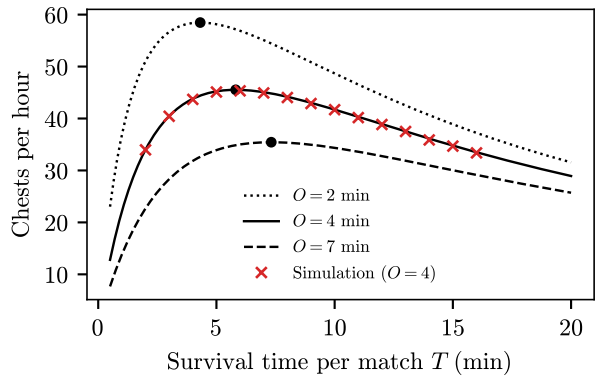


Figure 4: Chests per hour against survival time per match, for three values of the overhead O. Dots mark T^* from Proposition 1; crosses are the full simulation at $O = 4$.

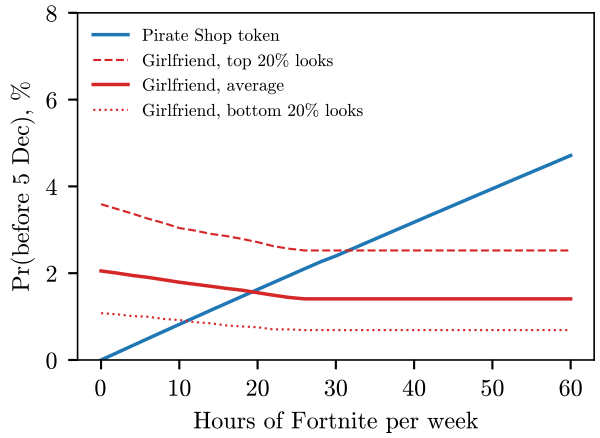


Figure 5: Probability of each event before the season ends on 5 December 2026, by weekly hours of Fortnite and the man's looks percentile.

4.7 The long run

If the token stayed in the game forever, the race between the two events would be even at about 14 hours a week (Figure 6). At 5 hours the girlfriend comes first 75% of the time (median 3.3 years against 12.3 for the token). At 30 hours the token comes first 68% of the time (2.0 against 4.7 years). The token wins at high hours mainly because the girlfriend has slowed down, not because the token has become likely.

5 Uncertainty

Many of the parameters above are assumptions. Rather than vary them one at a time, I give each of the 22 assumed parameters a uniform prior over a plausible range (Appendix C), draw 400 parameter sets, and rerun the entire pipeline for each one. That includes re-

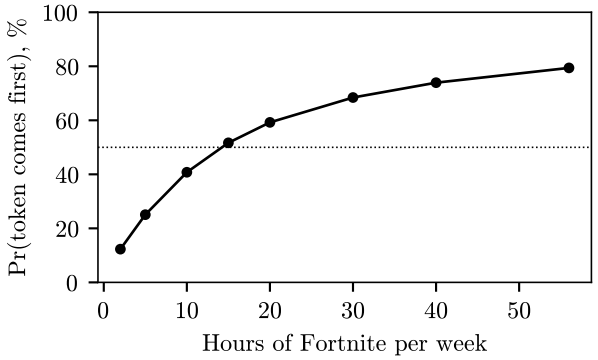


Figure 6: Probability that the token comes before a relationship, if the token never left the game.

Table 2: Headline results across 400 draws from the priors: median and 90% interval.

Quantity	Median	90% interval
Pr(token by 5 Dec), 10 h/wk	0.77%	0.50 – 1.19
Pr(girlfriend by 5 Dec), 10 h/wk	2.44%	1.56 – 3.66
Crossover, average man (h/wk)	25	15 – 46
Crossover, bottom 20% looks	14	8 – 23
Crossover, top 20% looks[†]	40	23 – 58
Time outside a match, 800th game	53%	42 – 64
Drake multiplier M	12	7 – 23

[†] Exists below 60 h/wk in 74% of draws.

calibrating M and ρ to the survey targets, which are themselves drawn between the Pew and GSS values for the single share and between 30% and 50% for the online share. Every draw could be calibrated.

Table 2 gives the results. At 10 hours a week, a girlfriend is more likely than the token in all 400 draws. The crossover moves around a good deal, from 15 to 46 hours a week for the average man, but it exists below 60 hours in 98.5% of draws. The baseline estimate of 19 hours sits in the lower part of this range, because the priors allow for breakup rates and single shares that make relationships easier to find. Figure 7 shows the bands.

To see which assumptions drive the answer, I estimate first-order variance-based (Sobol) sensitivity indices, $S_i = \mathrm{Var}(\mathbb{E}[Y \mid X_i])/\mathrm{Var}(Y)$, by binning each parameter into deciles. For the crossover (Figure 8), four parameters explain about 70% of the variance: the maximum number of chests per game ($S = 0.26$), the breakup rate (0.20), the looting time constant (0.17) and the single-share target (0.09). Everything about dating apps, DMs and swiping has an index below 0.05, because the calibration absorbs it. If someone wants to argue with the result, the most useful thing they could measure is how many chests a Fortnite player opens per game.

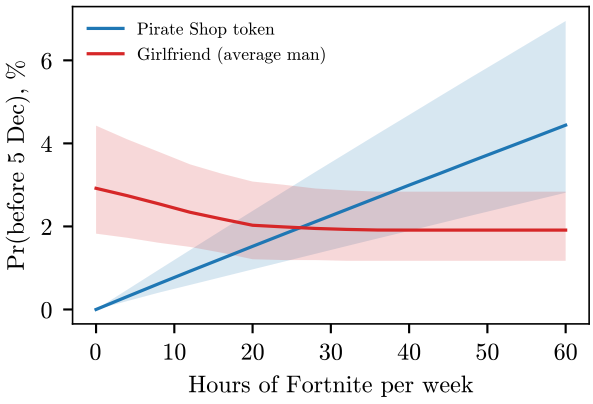


Figure 7: Season race with 90% bands across 400 draws from the priors.

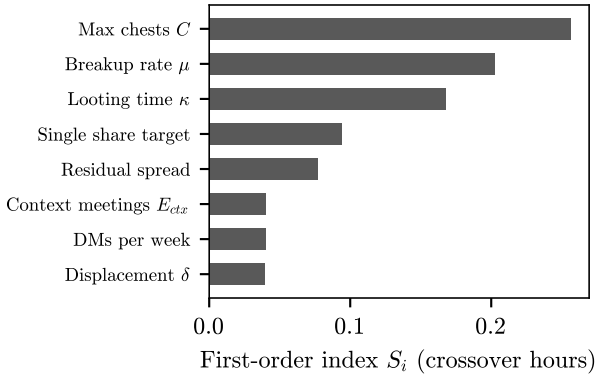


Figure 8: First-order sensitivity indices for the crossover point, top eight parameters.

6 Limitations

Chest counts and queue times are not published, so the looting and overhead parameters are informed guesses. The fatigue estimate comes from League of Legends, not Fortnite. The Tinder match rate is from London in 2016 and may be worse now. The relationship data are American and describe all young men, not Fortnite players. Gaming time is not randomly assigned, so the displacement effect is not causal, and the looks model compresses a complicated thing into one number, which is also what dating apps do. Finally, I treat a relationship as a single event, although it is followed by a second process with its own hazard rate, which I leave to future work.

7 Conclusion

Per attempt, love beats loot on every channel. Per hour, it depends on how a man spends his hours, what he looks like, where he lives and whether he swipes in the queue. For the average man at the ATUS average, a girlfriend remains the more likely outcome this season, though neither is likely. Above about 19 hours a week, the free Item Shop takes the lead. For the reader wondering which line on Figure 5 is theirs, I note that only one of the variables on it is under their control.

Acknowledgements

I thank the players in my lobbies, who supplied most of the data on early deaths.

Code and data availability

All code, parameters and results are at `github.com/djangothompson12-alt/Fortnite-Token-Probabilit` `python run_all.py` reproduces every number in the main text, and `python psa.py` reproduces Section 5.

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A Simulation algorithm

Algorithm 1 Time to token for player i

draw skill $z_i \sim \mathcal{N}(0, 0.35^2)$; $T_i^{\infty} \leftarrow 6e^{z_i}$
draw $K_i \sim \text{Geometric}(p)$ ▷ chests until token
$n \leftarrow 0$; $c \leftarrow 0$; $h \leftarrow 0$
while $c < K_i$ **do**
  $k \leftarrow n \bmod n_{\text{session}}$
  $\bar{T} \leftarrow (1-\phi)^k[T_i^{\infty} + (T_0 - T_i^{\infty})e^{-n/\tau}]$
  $T \sim \text{Gamma}(2, \bar{T}/2)$
  $c \leftarrow c + \text{Poisson}(\sigma_m C(1 - e^{-(1-\phi)^k T/\kappa}))$
  $h \leftarrow h + (T + O_n)/60$; $n \leftarrow n + 1$
end while
return h

Relationship times are drawn as $t_j \sim \text{Exp}(\lambda_j)$ with λ_j from Eq. (4). The calibration solves for M (inner root-finding, Brent’s method) and ρ (outer) on a fixed population of 200,000 simulated men.

B Monte Carlo error

At 10 hours a week, the season probability of the token is $0.8200\% \pm 0.0003$ percentage points (one standard error, 20,000 players, using the conditional estimator) and the probability of a relationship is $1.70\% \pm 0.04$ points (100,000 men). Figure 9 shows the running estimate. Monte Carlo error is small next to the parameter uncertainty in Section 5, so differences of a few hundredths of a percentage point between tables reflect different random seeds, not different models.

C Priors

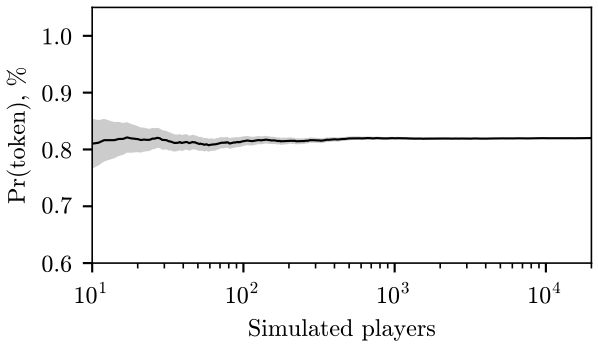


Figure 9: Running Monte Carlo estimate of the token’s season probability at 10 hours a week, with 95% band.

Parameter	Symbol	Uniform range
Max chests per game	C	8 – 16
Looting time constant	κ	3 – 10 min
Base overhead	O_0	2.5 – 6 min
Queue growth	Δ_Q	0 – 3 min
Fatigue per match	ϕ	0 – 0.02
Survival floor (median)	T^{∞}	4 – 9 min
New-account survival	T_0	11 – 18 min
Skill spread (log-sd)		0.2 – 0.5
Breakup rate	μ	0.25 – 0.5 /yr
Residual spread (log-sd)		0.25 – 1.0
Displacement	δ	0.1 – 0.4
Meeting rate per social hour	β	0.4 – 1.6
Context meetings per day	E_{ctx}	0.3 – 1.0
Swipes per day		10 – 100
Match → conversation		0.15 – 0.5
Conversation → date		0.08 – 0.25
Cold DMs per week		0 – 10
DM reply rate		0.03 – 0.15
Looks range, online		4 – 20
Looks range, in person		1.5 – 6
Single share target	s	0.53 – 0.63
Online share target		0.30 – 0.50